Bank Account Withdrawal System

Category B Practice - Part A, Problem 1

Problem summary

Savings accounts must maintain at least 1000; Current accounts may go to -5000. Read operations line by line. For each WITHDRAW, print the result, reason for rejection, or Account not found.

Java solution

```
import java.util.*;
public class Main {
 abstract static class Account {final String id;double balance;Account(String i,double b){id=i;balance=b;}
  abstract boolean allowed(double a);abstract String reason();
  boolean withdraw(double a){if(a<=0||!allowed(a))return false;balance-=a;return true;}}
 static class Savings extends Account {Savings(String i,double b){super(i,b);}boolean allowed(double a){return b
 static class Current extends Account {Current(String i,double b){super(i,b);}boolean allowed(double a){return b
 static String fmt(double n){return n==Math.rint(n)?Long.toString((long)n):Double.toString(n);}
 public static void main(String[] args){Scanner sc=new Scanner(System.in);Map<String,Account> m=new HashMap<>();
  while(sc.hasNextLine()){String line=sc.nextLine().trim();if(line.isEmpty())continue;String[] p=line.split("\\s
   if(p.length==3&&p[0].equals("Savings"))m.put(p[1],new Savings(p[1],Double.parseDouble(p[2])));
   else if(p.length==3&&p[0].equals("Current"))m.put(p[1],new Current(p[1],Double.parseDouble(p[2])));
   else if(p.length==3&&p[0].equals("WITHDRAW")){Account a=m.get(p[1]);double amount=Double.parseDouble(p[2]);
    if(a==null)System.out.println("Account not found");
    else if(a.withdraw(amount))System.out.println(a.id+" balance "+fmt(a.balance));
    else System.out.println(a.id+" rejected: "+(amount<=0?"amount must be positive":a.reason()));
   }
  }
 }
}
```

Example input

```
Savings S1 5000
WITHDRAW S1 4500
Current C1 2000
WITHDRAW C1 6000
WITHDRAW S1 3000
WITHDRAW C1 2000
```

Expected output

```
S1 rejected: minimum balance 1000
C1 balance -4000
S1 balance 2000
C1 rejected: overdraft limit 5000
```

Notes

Average account lookup with HashMap is O(1). A rejected withdrawal does not change the balance.